Quiz 9

1. Show that the following identity holds:

$$\frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta} = 2\csc\theta$$

$$\frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta}$$

$$= \frac{\sin\theta(\sin(\theta)) + (1+\cos\theta)\cdot(1+\cos\theta)}{(1+\cos\theta)\cdot\sin\theta}$$

$$= \frac{\sin^2\theta + 1 + 2\cos\theta + \cos^2\theta}{(1+\cos\theta)\cdot\sin\theta} = \frac{1+1+2\cos\theta}{\sin\theta\cdot(1+\cos\theta)} = \frac{2(1+\cos\theta)}{\sin\theta(1+\cos\theta)}$$

$$= \frac{2}{\sin\theta} = 2\cdot\csc\theta$$

2. Show that the following identity holds:

$$\frac{\tan\theta + \cot\theta}{\sec\theta\csc\theta} = 1$$

$$= \frac{\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta}}{\sec\theta \cdot \csc\theta} = \frac{\frac{\sin^2\theta + \cos^2\theta}{\cos\theta\sin\theta}}{\sec\theta \cdot \csc\theta} = \frac{1}{\sin\theta\cos\theta} \cdot \frac{1}{\sec\theta\csc\theta}$$

$$= 1$$